

Cubic interpolation

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Let the input data be defined as a set of x and y points on full levels (Table 1). Let us assume that $x_{i+1} - x_i$ is a constant, Δx . Cubic interpolation will be performed between (full) points x_3 and x_4 (bold in the table) using the Stirling and Bessel formulae (below, also see [1], Eqs. (5-3.5) and (5-3.6)).

- For interpolation to points within this domain that are closer to x_3 than $x_{3.5}$ or x_4 , Stirling formula 1 is used (reference point x_3):

$$y(x_3 + \delta x) = y_3 + y'_3 \delta x + y''_3 \frac{\delta x^2}{2} + y'''_3 \frac{\delta x (\delta x^2 - \Delta x^2)}{6}.$$

- For interpolation to points within this domain that are closer to x_4 than $x_{3.5}$ or x_3 , Stirling formula 2 is used (reference point x_4):

$$y(x_4 + \delta x) = y_4 + y'_4 \delta x + y''_4 \frac{\delta x^2}{2} + y'''_4 \frac{\delta x (\delta x^2 - \Delta x^2)}{6}.$$

- The remaining points within this domain are closer to $x_{3.5}$ than x_3 or x_4 , the Bessel formula is used (reference point $x_{3.5}$):

$$y(x_{3.5} + \delta x) = y_{3.5} + y'_{3.5} \delta x + y''_{3.5} \frac{\delta x^2 - \frac{\Delta x^2}{4}}{2} + y'''_{3.5} \frac{\delta x \left(\delta x^2 - \frac{\Delta x^2}{4} \right)}{6}.$$

References

- [1] Cornelius Lanczos. *Applied analysis*. Dover Publications, NewYork, 1988.

input			interpolate		derivatives		
	x	y	x	y	dy/dx	d^2y/dx^2	d^3y/dx^3
Full	x_1	y_1	x_1	y_1			
Half			$x_{1.5}$	$y_{1.5}$			
Full	x_2	y_2	x_2	y_2	y'_2		
Half			$x_{2.5}$	$y_{2.5}$	$y'_{2.5}$	$y''_{2.5}$	
Full	x_3	y_3	x_3	y_3	y'_3	y''_3	y'''_3
Half			$x_{3.5}$	$y_{3.5}$	$y'_{3.5}$	$y''_{3.5}$	$y'''_{3.5}$
Full	x_4	y_4	x_4	y_4	y'_4	y''_4	y'''_4
Half			$x_{4.5}$	$y_{4.5}$	$y'_{4.5}$	$y''_{4.5}$	
Full	x_5	y_5	x_5	y_5	y'_5		
Half			$x_{5.5}$	$y_{5.5}$			
Full	x_6	y_6	x_6	y_6			

Table 1: Data table for cubic interpolation. The input x and y points are defined on full points, but the interpolation scheme will need interpolated information also on half points (equidistant between the full points). The numerical derivatives are computed based on centred differences.